Mobility in Socio-Technical Urban Organisms

from sensing to steering



Franco Zambonelli

Università di Modena e Reggio Emilia

franco.zambonelli@unimore.it

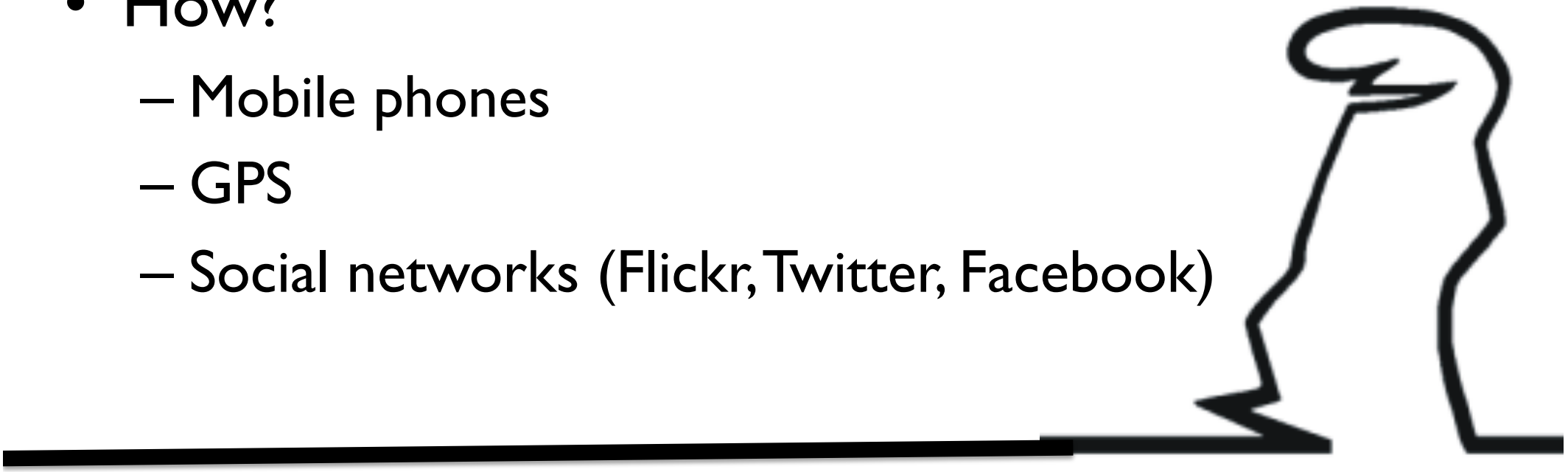
Outline

- Sensing Mobility
- Towards Urban Organisms
- Mobility in Socio-Technical Urban Organisms
- From Sensing to Steering
- Motivations and Examples
- The SAPERE Approach



Sensing Mobility

- Why?
  - Because now it can be done
  - To understand mobility patterns
  - To Improve pervasive services
- How?
  - Mobile phones
  - GPS
  - Social networks (Flickr, Twitter, Facebook)



Experiences in Sensing Mobility

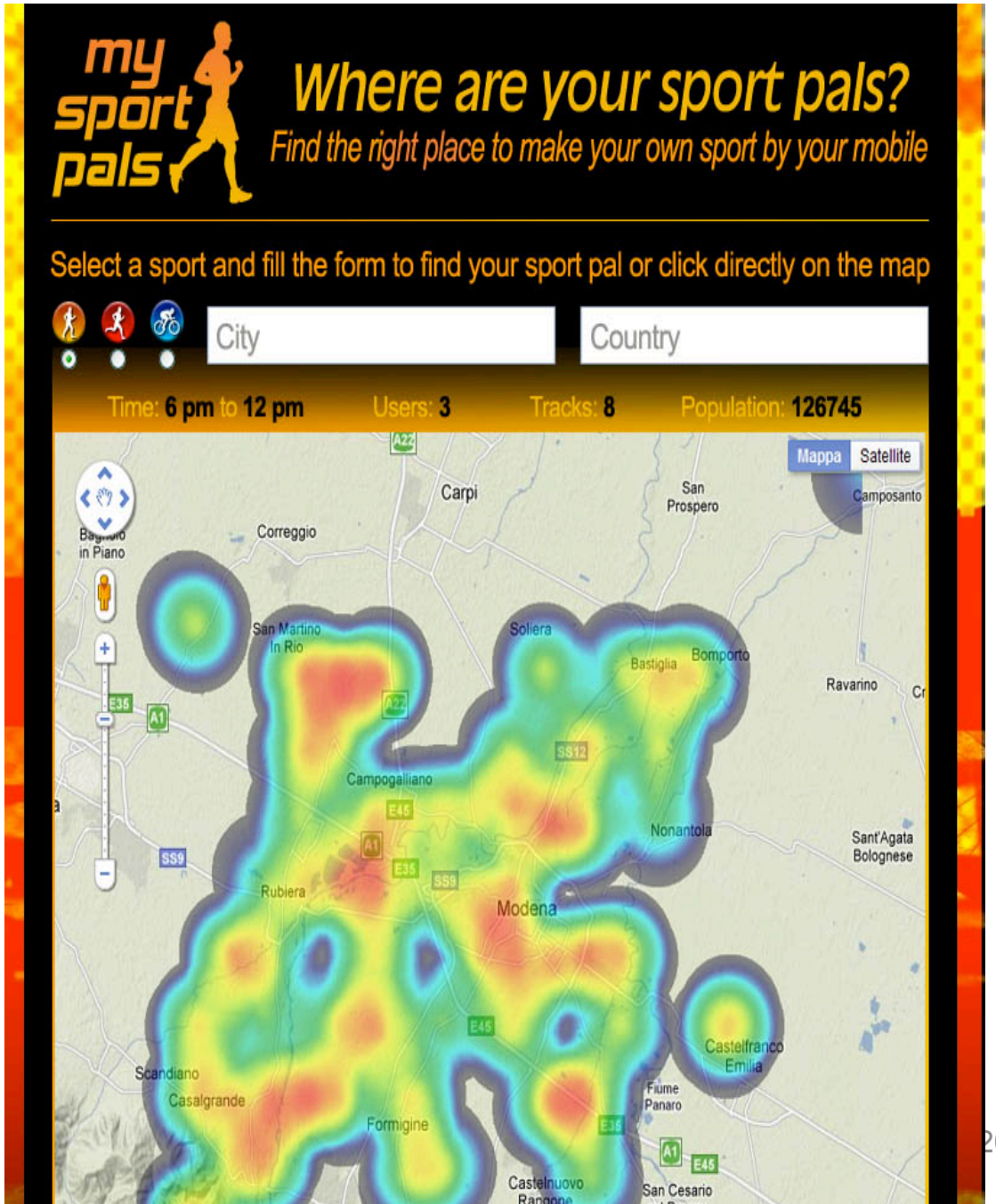
@ The Agents and Pervasive Computing Group

- Via GPS
  - Building diaries :: recognising activities
- Via Cell Phone Analysis
  - See following talk by Marco :: event recognition
- Via Social Networks
  - Nokia sports tracker, Flickr, Twitter…



Nokia Sport Tracker

Ferrari and Mamei, IEEE Computer, 2011



- See where people moves (e.g., bikes) in a city
  - Recurrent patterns via Kernel Density Estimation
- To understand
  - How people uses public spaces
  - How such public spaces could be re-engineered
  - E.g., plan for new bicicle paths
- See
  - http://www.mysportpals.com/

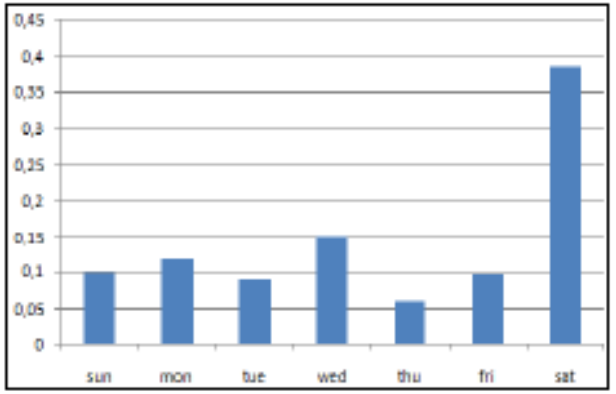
Twitter as a Mobility Sensor

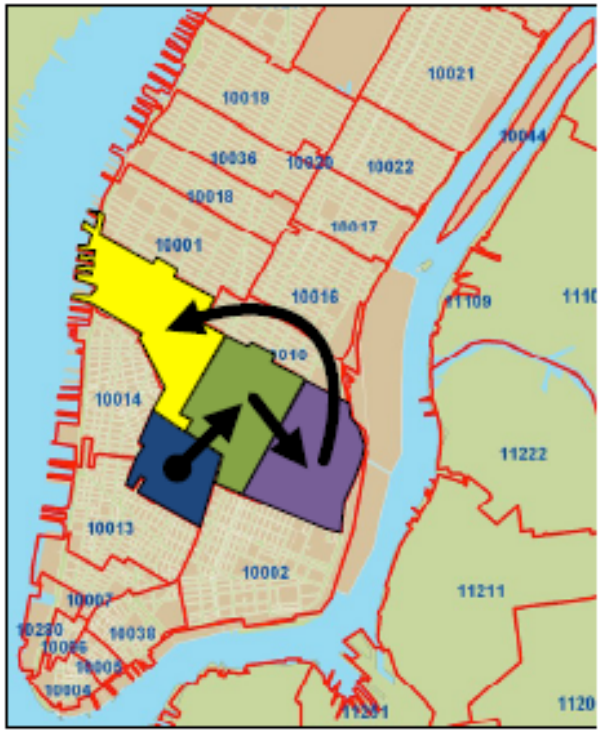
Ferrari et al 2011, ACM Sigspatial LBSN, Chicago, June 2011



- Get all geotagged Tweets in a city
- Apply summarizing techniques (LDA)
- Exctract recurring crowd behaviors
  - Where do people go on Sat night?
  - Where first?
  - Where last?

Topic 4 (top words)		
zip code	hour	prob
10011	0:00-2:00	0,35
10003	20:00-22:00	0,15
10012	18:00-20:00	0,08
10009	22:00-24:00	0,02





Flickr as a Recommender

Rosi et al 2010, IE '10 Conference, Singapore, June 2010

- Mining Flickr geotagged pictures
  - Recognize paths of tourists and hot spots
  - Build "typical" paths around the city
- Then, for different categories of user profiles
  - Suggest tourist where to go based on similarity

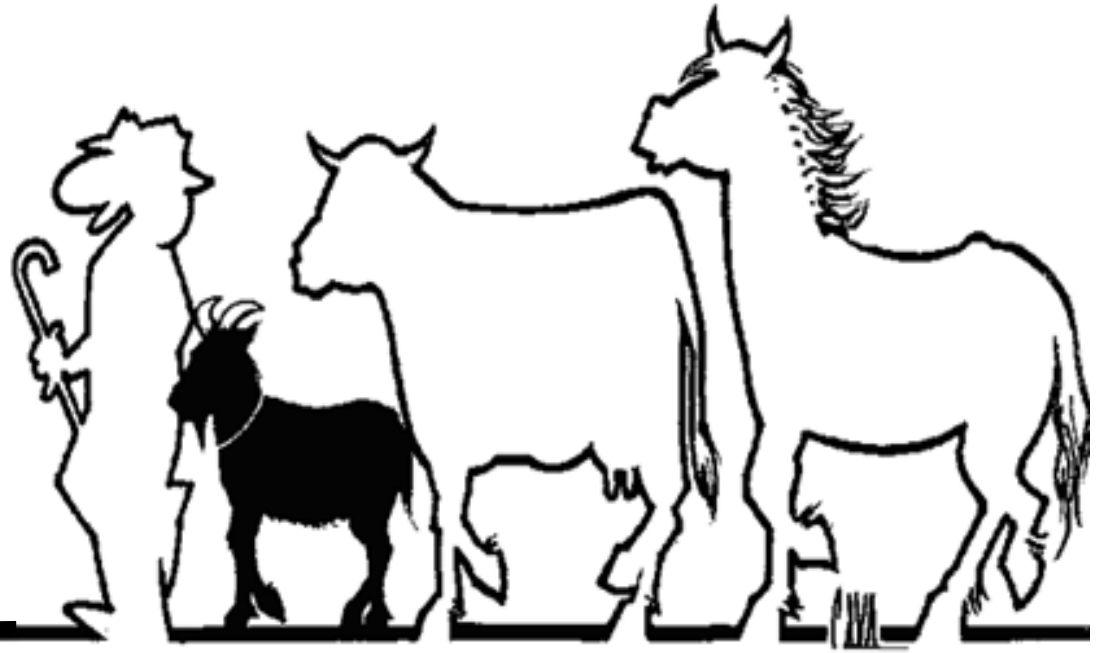




	London		Paris	
	Big Ben	Trafalgar Sq.	N. Dame	A. Triomphe
Alberto	yes	yes	yes	no
Marco	no	no	yes	yes
Franco	yes	yes	?	?

From Sensing to Steering

- Sensing mobility in urban scenarios is becoming common
- But our urban environments are becoming more than senseable
  - They can be rich in computing and actuating capabilities
  - There could be Human and ICT put at work in a coordinated way
- ICT and Humans can collectively work as an **urban superorganisms**



Urban Organisms Scenario: ICT Side

- An ICT-enriched urban environment with rich sensing, actuating, and computing (SAC) capabilities
  - **Sensing**: sensor networks, tags, smart objects,etc.
  - **Actuating**: traffic controllers, public digital displays, critical infrastructures
  - **Computing**: highly distributed and decentralized, with inter-connected computational engines everywhere



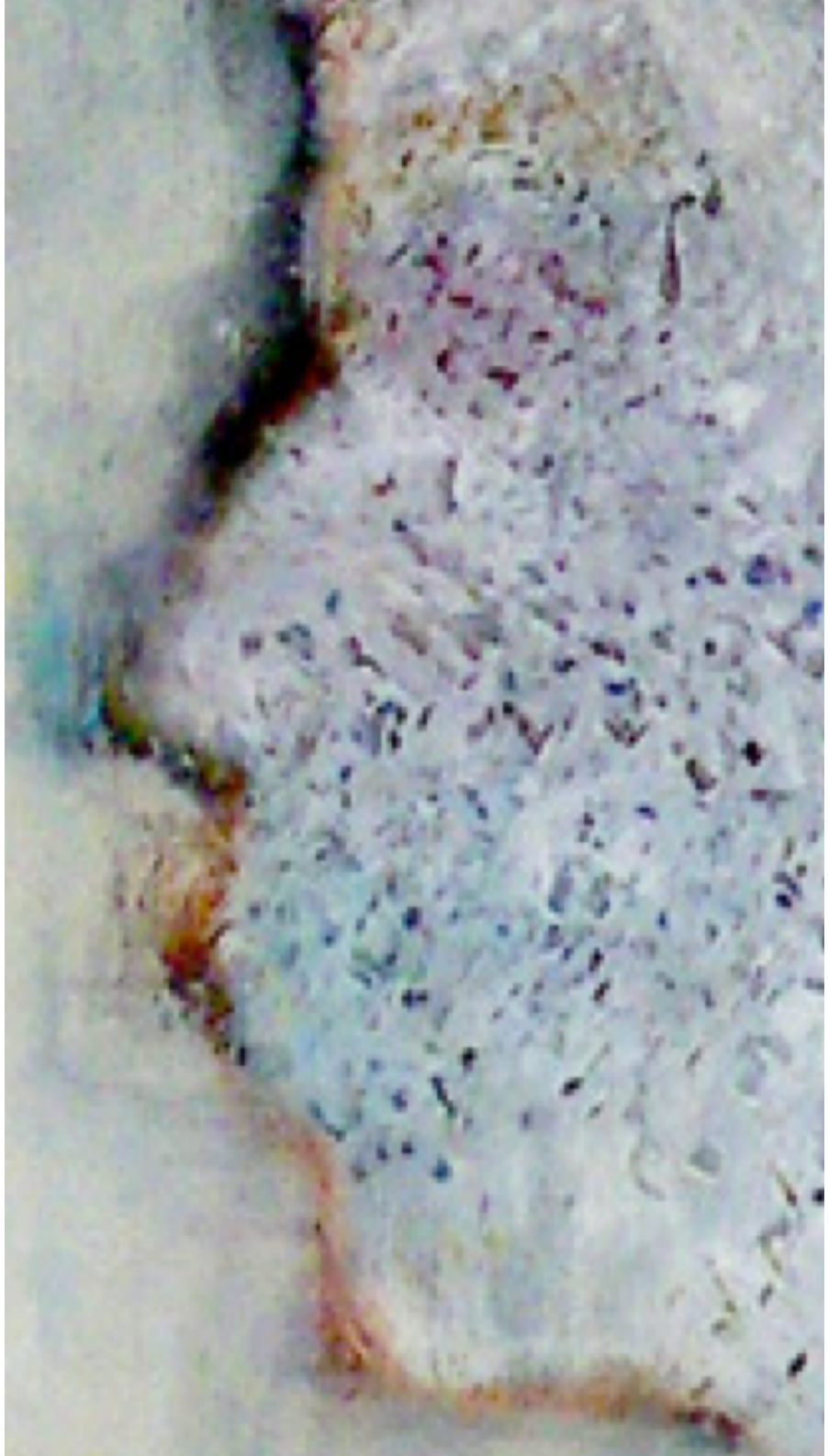
Urban Organisms Scenario: Human Side

- People with smart phones or alike (or whatever will appear in the future as wearable devices) contribute to such SAC capabilities
  - **Sensing**: the 5 senses + smart phones
  - **Actuating**: the body
  - **Computing**: human & social intelligence



Urban Organisms: Putting All Together

- The ICT and Human/Social level blurred to the point of invisibility:
- Their capabilities well complement each other in a process of extremely high value co-creation
- In the resulting overall "urban organism", we can achieve very high-levels of collective "urban" intelligence
  - That can dramatically change the way we **MOVE**, live, work, move, and play, in our towns



Steering Mobility in Urban Organisms

- **Mobility** *per se* :: car, bike, ride sharing
- **Crowdsorced city maintainance** :: go there do that
- **Exhibitions** :: avoid crowd or suggest paths
- **Support** :: guide children or tourists via displays
- All of these require
  - sensing, computing (data interpretation) actuation (steering)
  - **Adaptive self-organized mobility strategies**

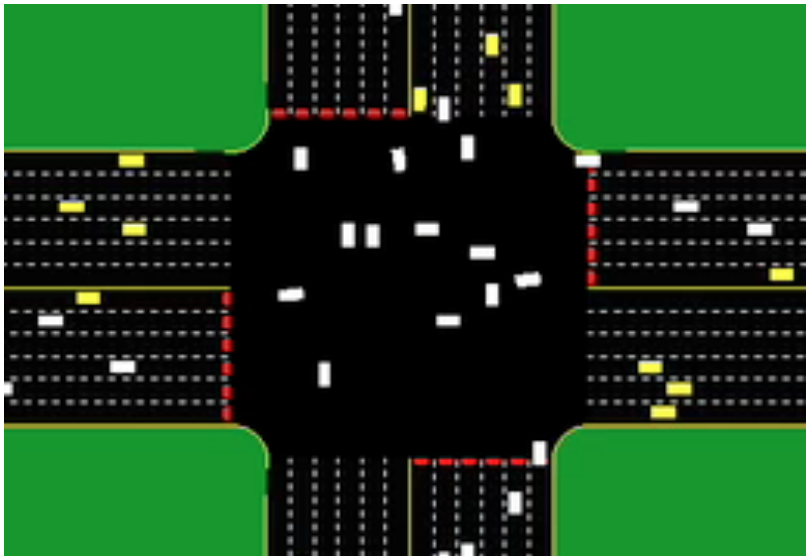


Example: Traffic Steering

- Typical solution to steer traffic: traffic lights
  - Not adaptive, not fault tolerant
- Self-organized human solutions
  - Crazy, but possible
- Self-organized ICT-based solutions
  - Seems less crazy, but will be possible in 5 years
- Integrating the two?







The SAPERE Project

- SAPERE "Self-aware Pervasive Service Ecosystems"
  - EU FP7 FET
  - Starting October 1st 2010, lasting 3 years
  - UNIMORE (Coordinator), UNIBO, UniGeneve, UniStAndrews, UniLinz
- Key objectives
  - To define and implement aframework for adaptive pervasive service ecosystems
  - Experience with pervasive urban services
  - There included self-organized crowd steering via pervasive displays









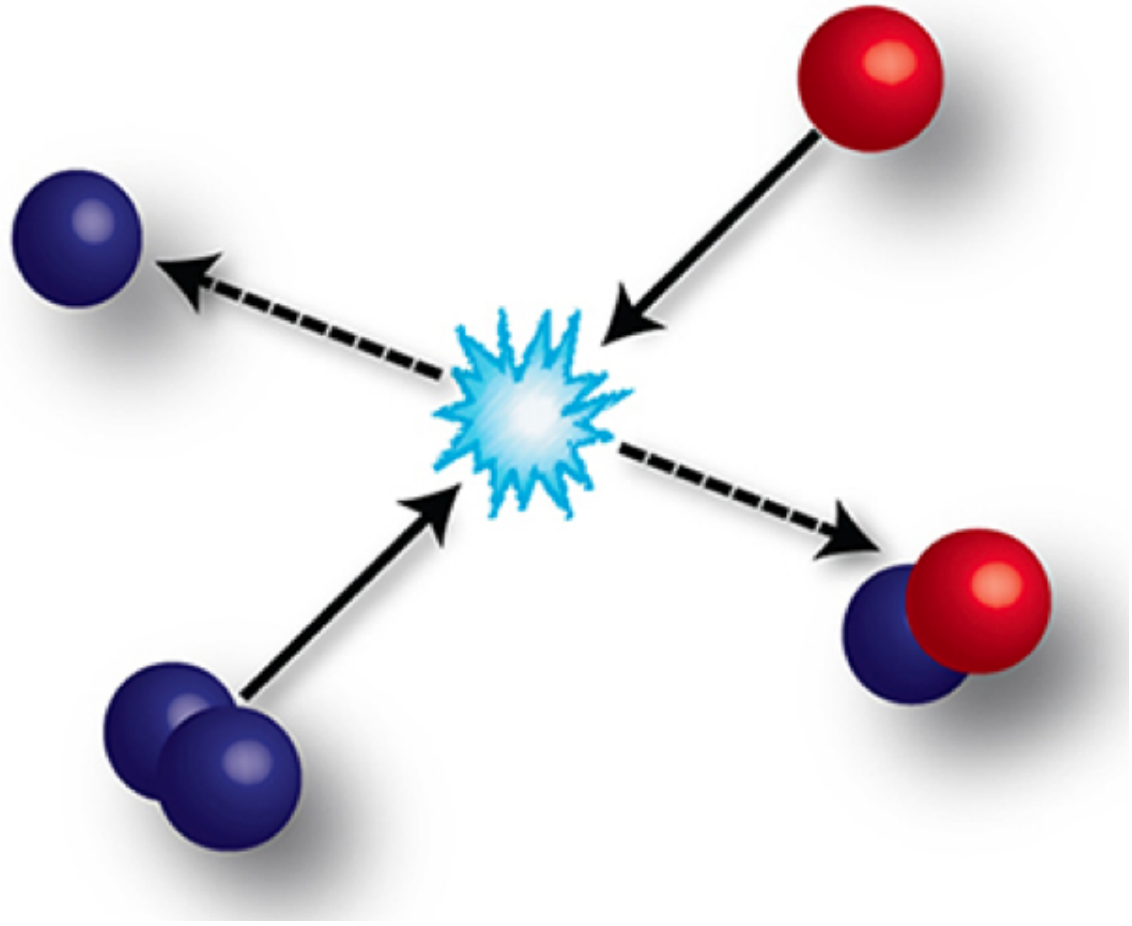






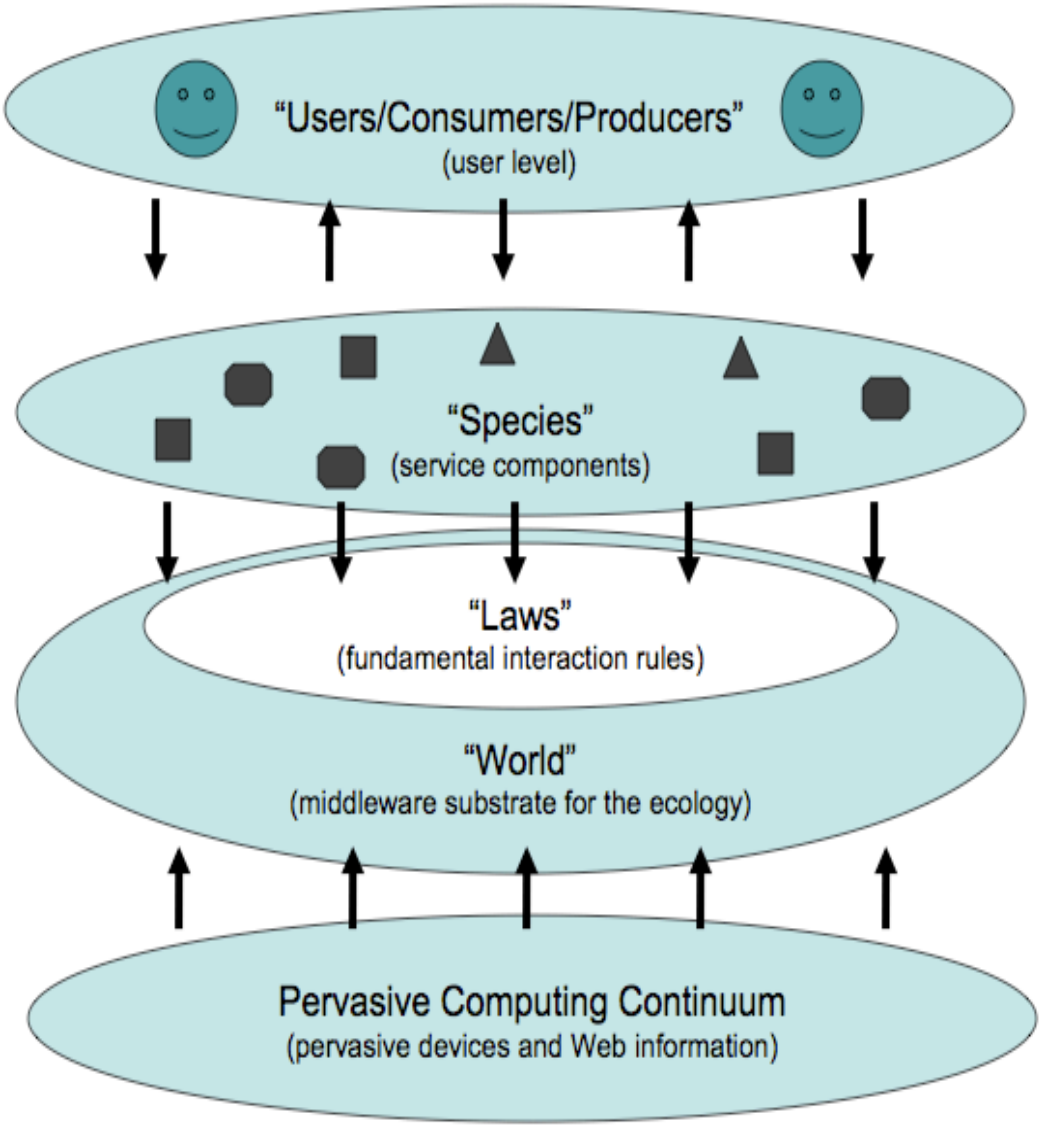
The SAPERE Approach

- Nature-inspired (Biochemical)
  - Simply metaphor for combining/aggregating services in a spontaneous way
- Spatially-situated
  - To match the nature of pervasive services (and of mobility issues)
- Self-organizing
  - Spontaneous patterns of services and users interaction



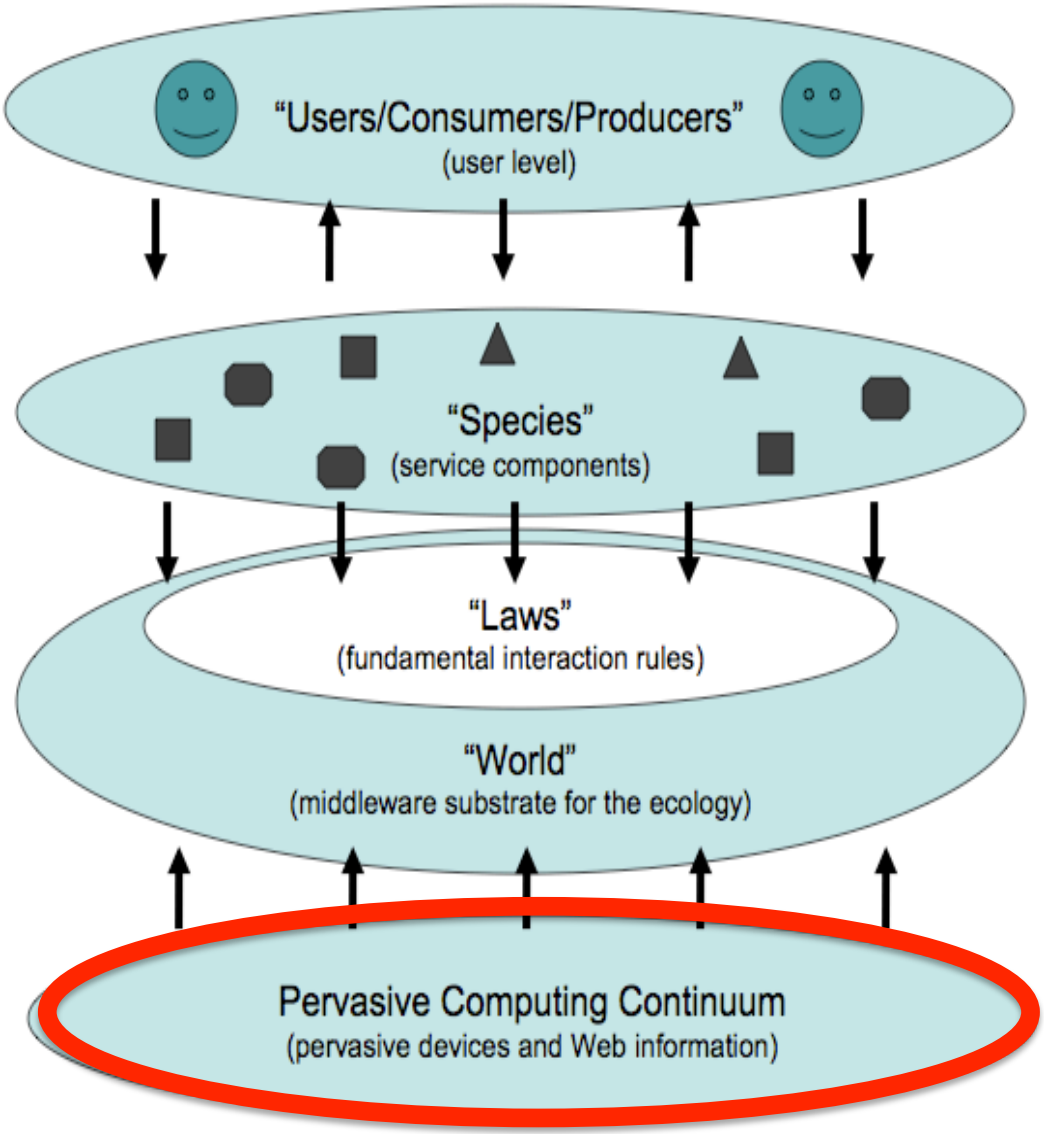
The SAPERE Reference Architecture

- It abstracts from any specific nature-inspired metaphor
  - (but we are committed to biochemical ones)
- Shows how general ecosystem concepts can be framed in a uniform way
- Useful to turn the architecture into an actual middleware infrastructure



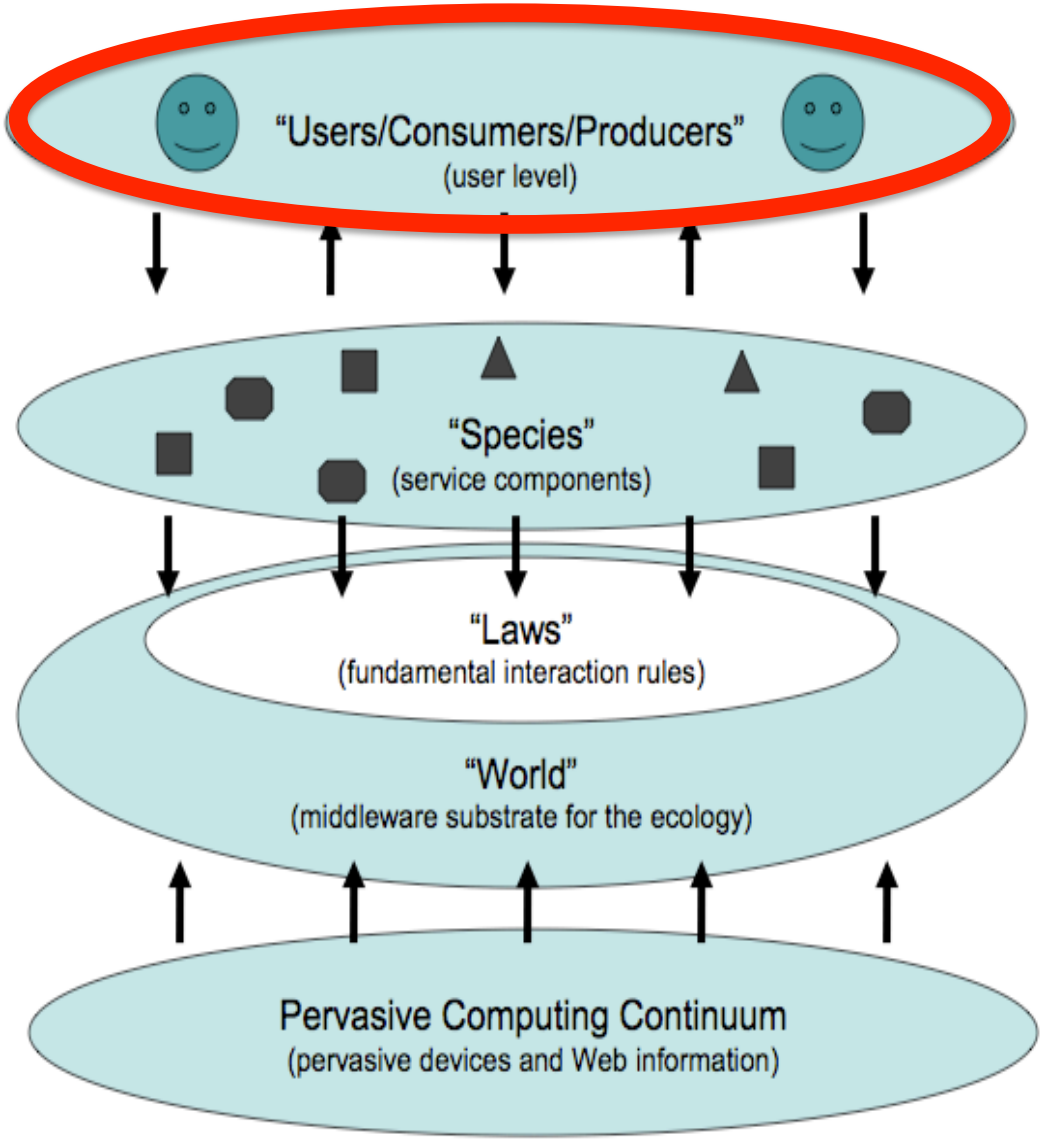
The Pervasive Computing Continuum

- Shaping the hardware ground on which the actual ecosystem will live and execute
  - Pervasive sensing and actuating devices very densely deployed in space
  - Personal computer-based systems
  - Wireless communications
- Feeding the ecosystem with data about nearly every facts of the world
  - Also via Web information



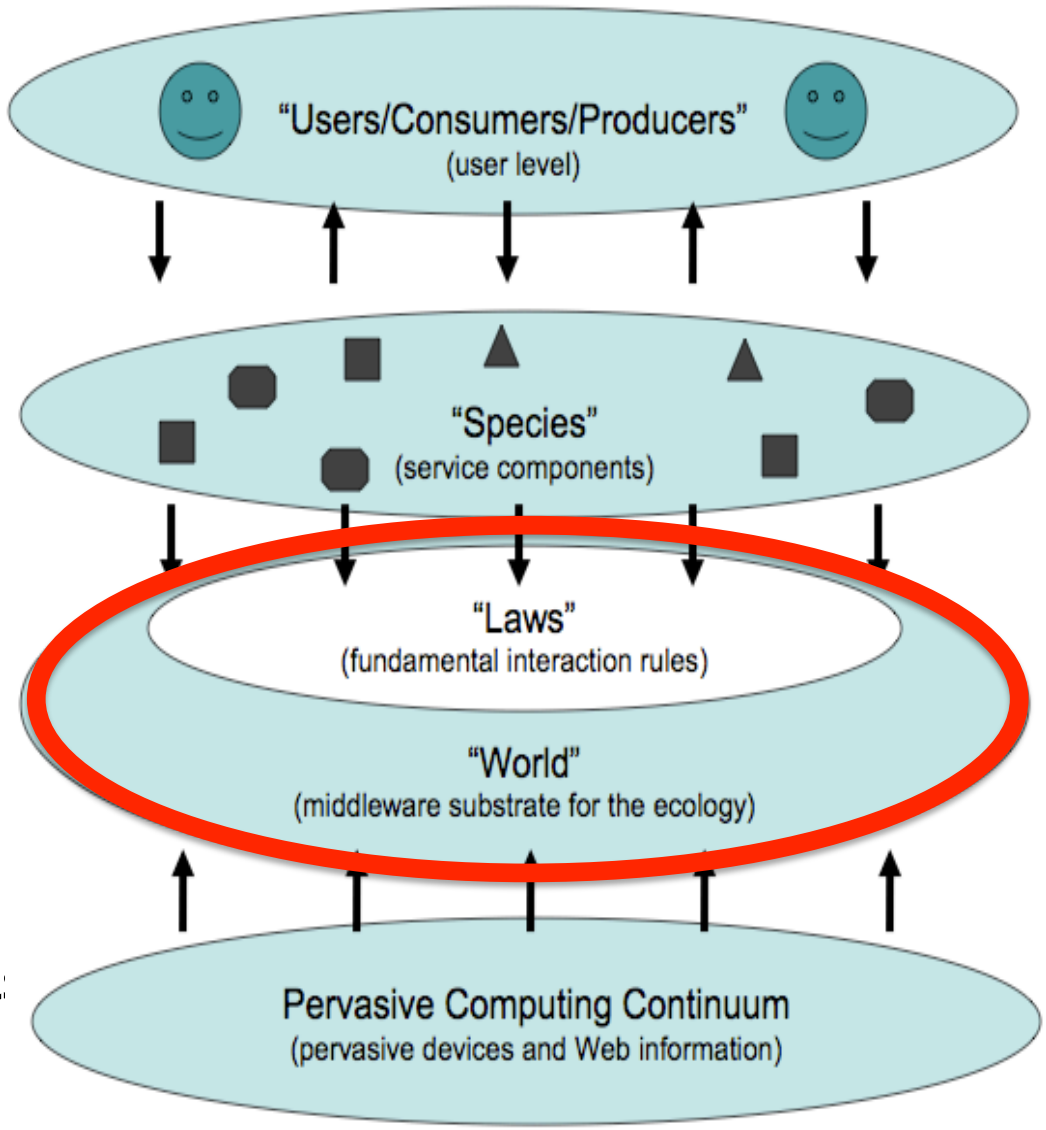
The SAPERE Users

- They can "*observe*", i.e., query, the ecosystem and its components
  - To obtain data, or results
  - In a fully decentralized way
- They can "*extract*" components
  - To consume data and service
- They can "*inject*" new components and data items
  - To personalize the network
  - To deliver own services
  - To enforce control



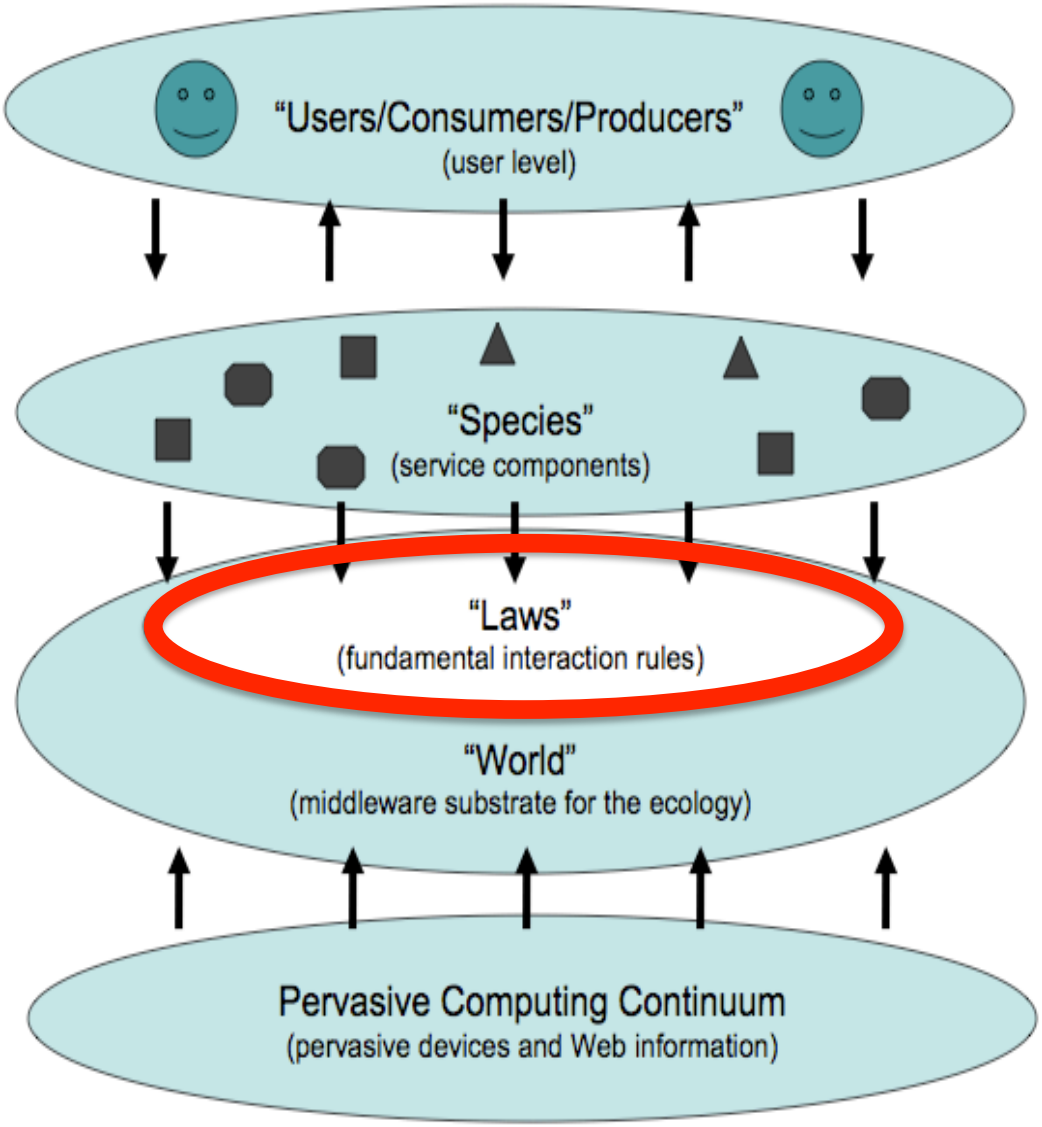
The Spatial Middleware

- A very minimal middleware substrate
  - No "smart" middleware services
  - Networked reactive tuple spaces
- Key goals
  - Supporting the spatial lifecycle of components over a dynamic substrate
  - Enabling and enforcing interactions across components
  - According to the "laws of nature" of the ecosystem



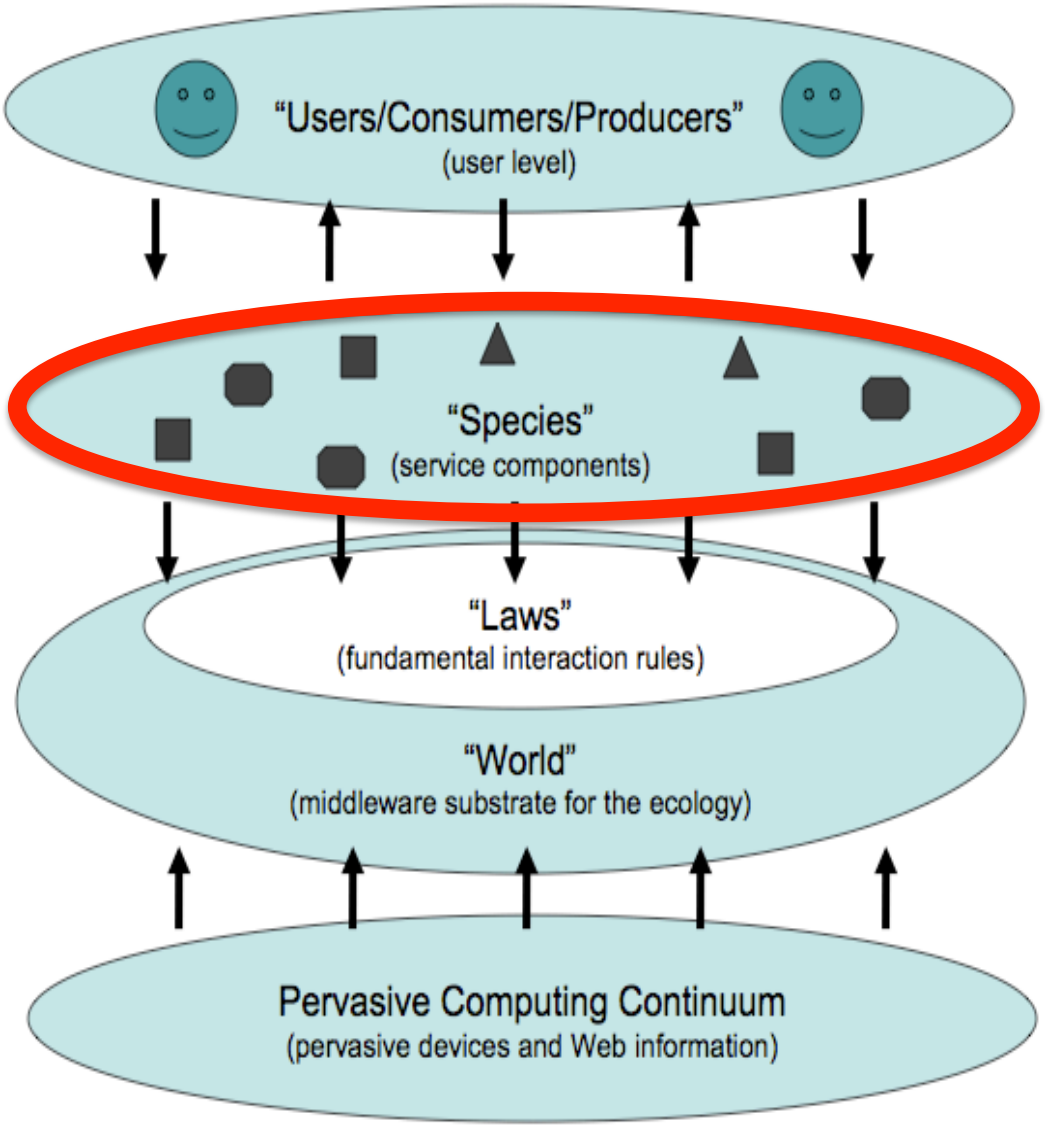
The Bio-Chemical Eco-Laws

- Ruling interactions and the overall dynamics and self-* behaviour of the system
  - How components should interact ("bond") and when
  - How components should compose ("catalysis")
  - How information should spread ("diffusion")
- They are eternal
  - Species of components can change, laws can't
  - Different species may react to laws in differentiated ways



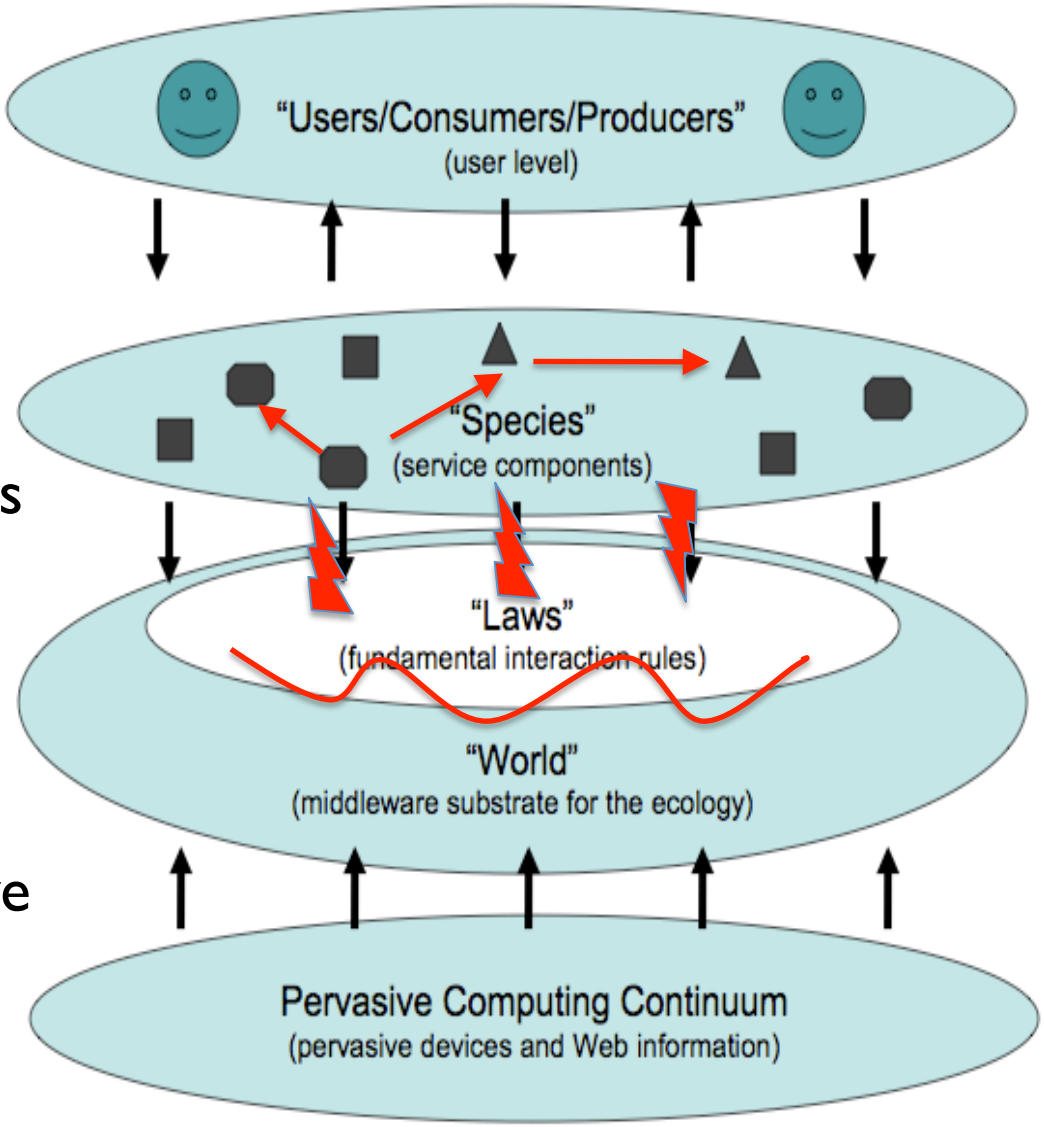
The SAPERE Species

- The software/digital components of the ecosystem
  - Sort of virtual molecules
  - Software agents in the end
- May be of different nature
  - Passive vs active
  - Decentralized production
- Are all subject to the laws
  - But different components can react differently to laws (as molecules do)
  - Based on internal characteritics and external interfaces



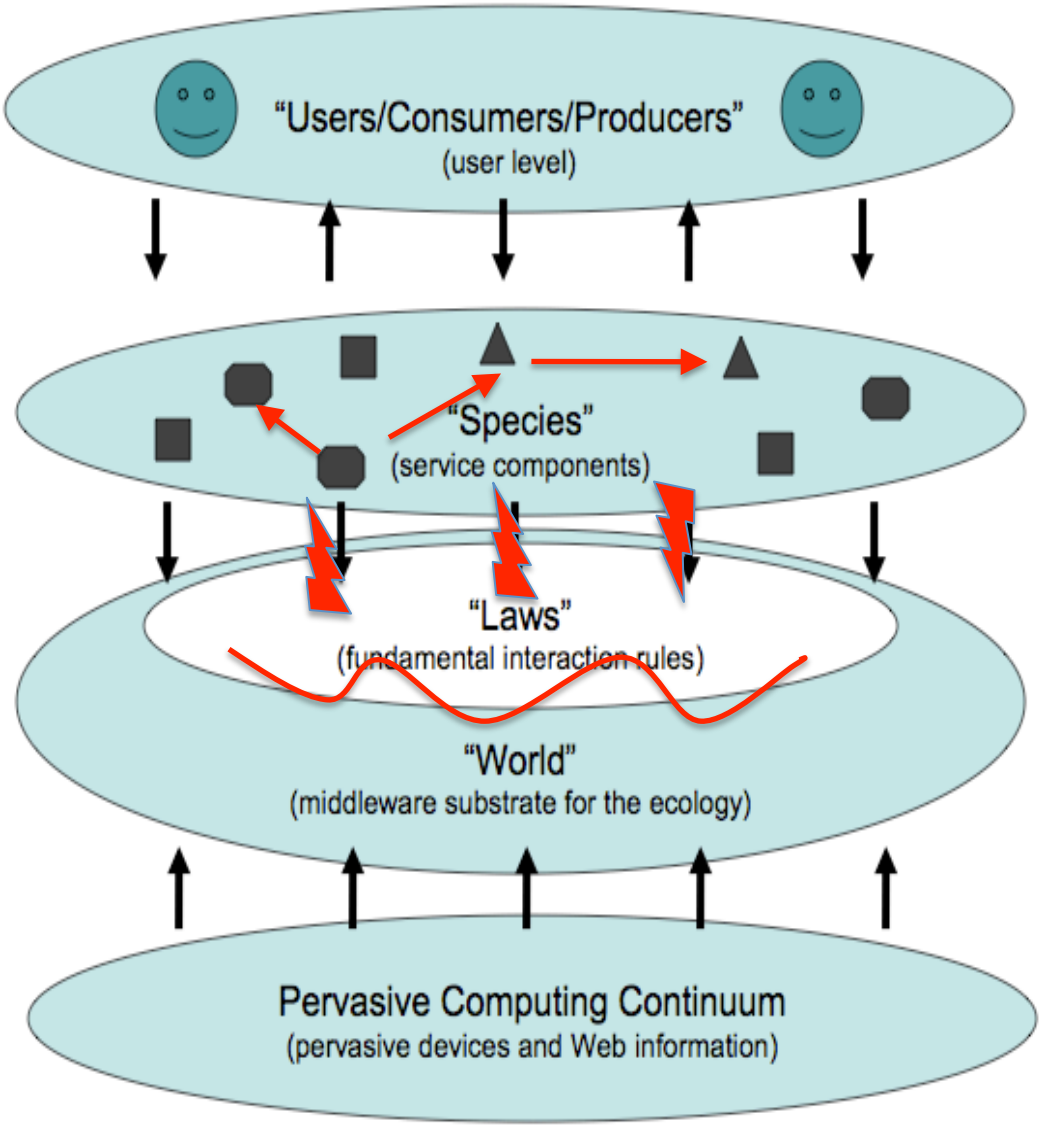
The Ecosystem Dynamics

- Species
  - Living in a region of "World"
  - Moving, acting, composing, as determined by laws
  - Not self-aware in theselves
- Laws
  - Local activities and interactions
  - Apply based on state of local components (feedback loops)
- World
  - The shape of space influence (and is influenced by) the above
- Dynamics
  - Seemingly self-aware system-level adaptability

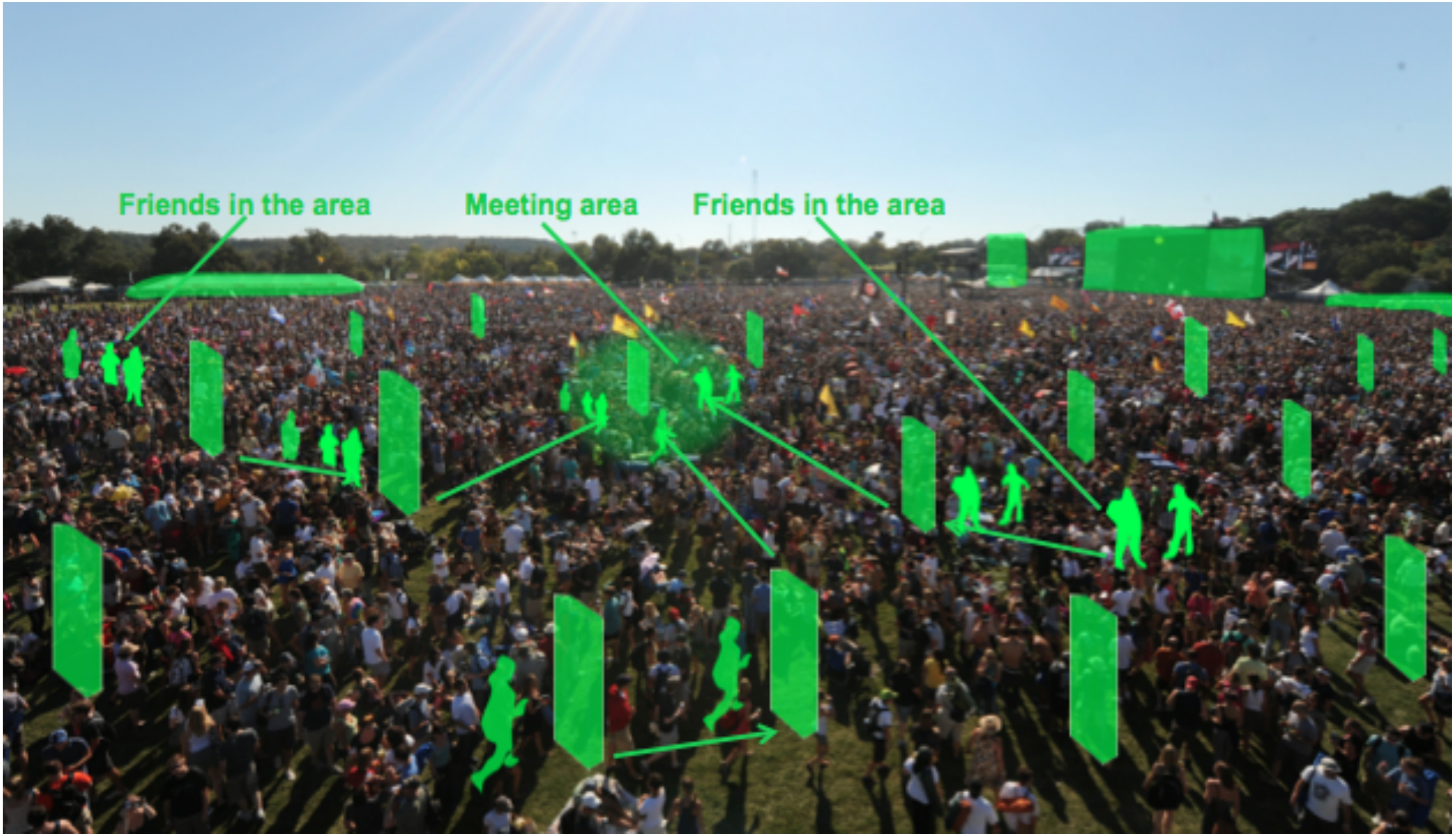


Where is Self-awareness?

- Self-awareness is not at the level of individual components
  - None of them is aware
  - They can only perceive and react to specific aspects of local spatial situation
- Self-awareness is a systemic, observable property of the ecosystem
  - Upon changes in a situation
  - There are spontaneous reactions in the ecosystems
  - As if it were aware of what's happening



Ecosystem of Displays: Scenario



Ecosystem of Displays: Use Cases

- Profle-based Adaptation





- Content follow users

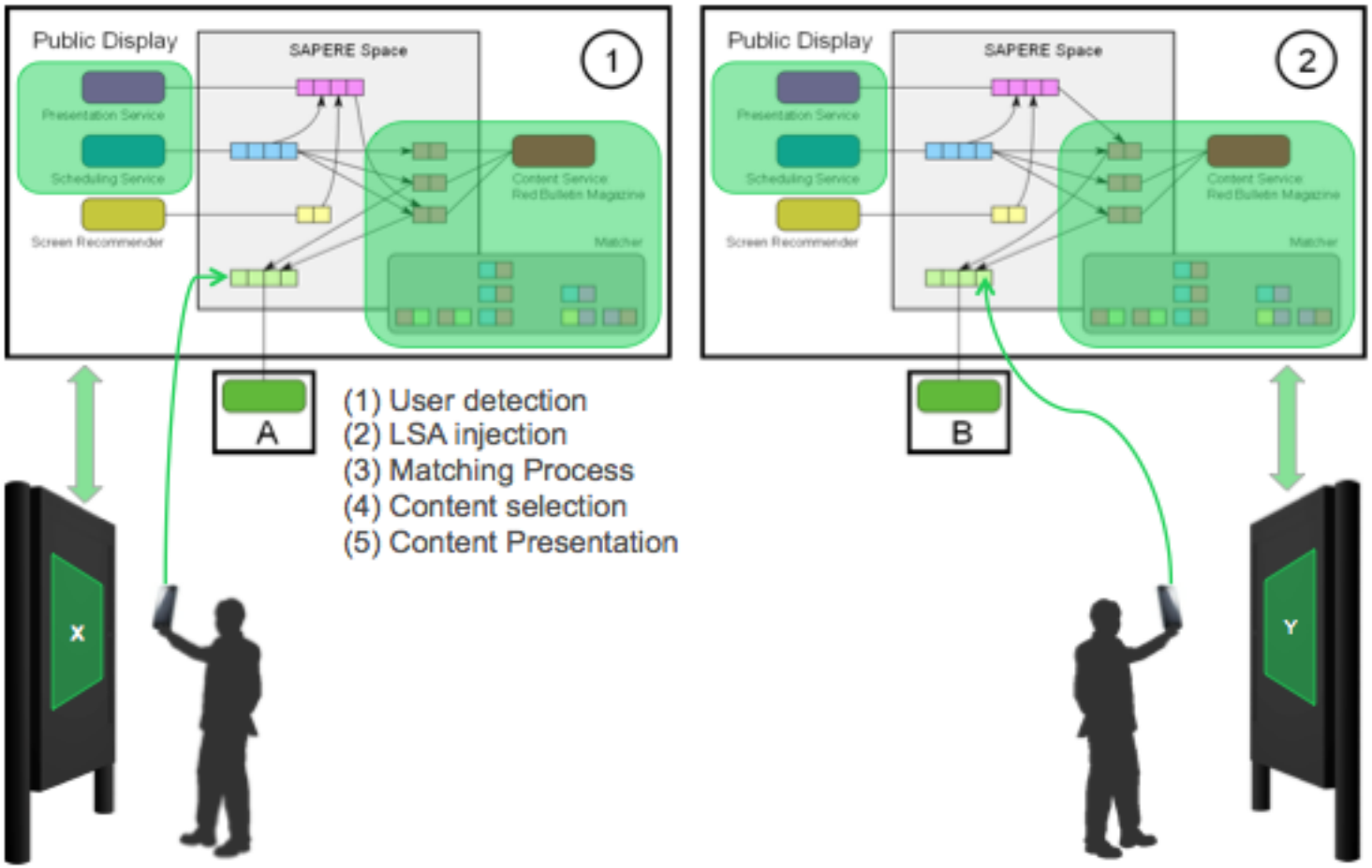
- Crowd steering



The Ecosystem of Displays: How it Works

– Users interact via sorts of chemical reactions

– Situatedness and spatiality



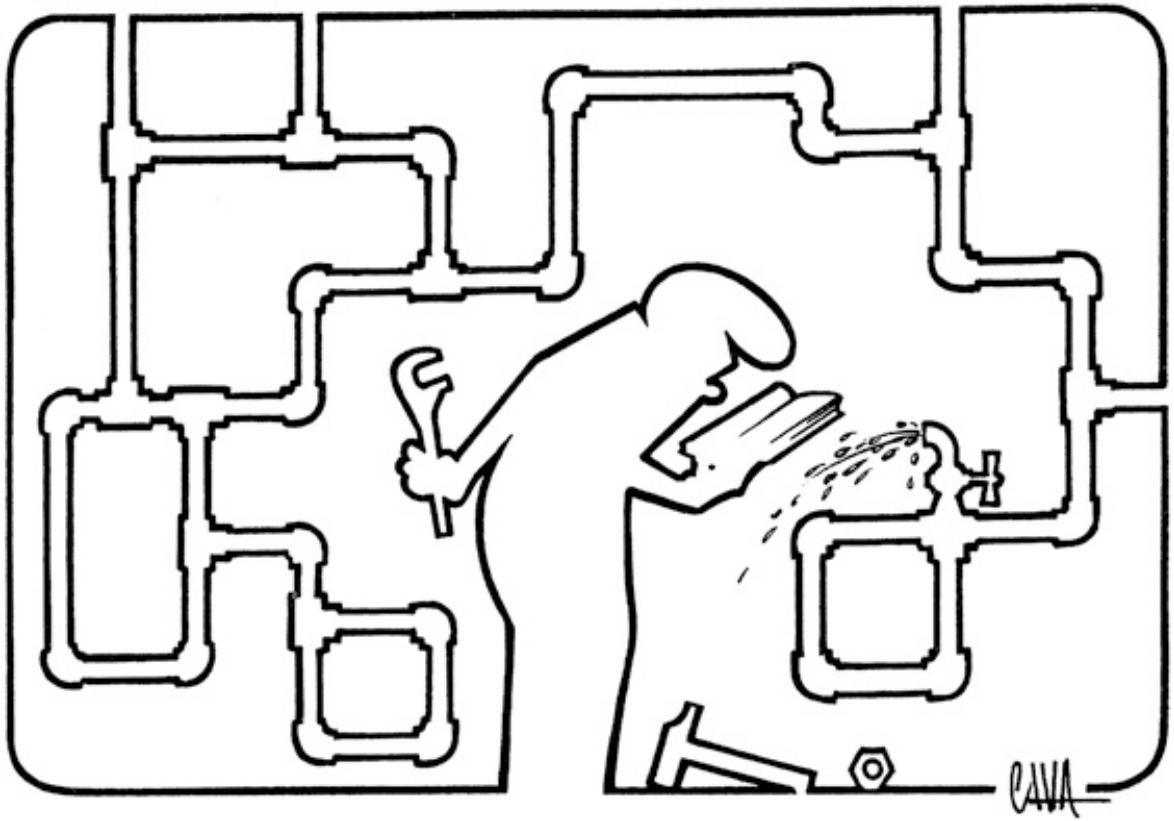
Crowd Steering in SAPERE



A SAPERE Demo by yhe Pervasive Computing Institute of JKU

Challenges

- Tools to engineer
  - What programming languages and abstractions?
  - Role of existing social networks in future ecosystems?
- Controlling emergent behaviors
  - What you steer is what you get?
- Incentives for participation
  - To change behavior
  - To accept being steered
  - Money? Badges? Reputation?



Conclusions

- Current focus on "**sensing mobility**" patterns
  - Good science to be make, yet…
- Pervasive computing technologies can be used to "**steer mobility**"
  - Affect urban mobility patterns
  - To adaptively regulate urban activities
  - To autonomously solve urban problems (maintenance, mobility, safety) as if the city were a sort of **superorganisms**
- SAPERE is build promosing models and infrastructures
  - And will test in pervasive displays for crowd steering
  - Yes there are a lot of challenges to solve, in general and beside SAPERE

